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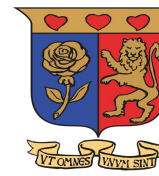
# Basic and Initial Switch Settings

A switch forwards traffic by default and does not need to be explicitly configured to operate. E.g. two configured hosts connected to the same switch can communicate. Regardless of the default behaviour of a new switch, all switches should be configured and secured.

| Console | This is a physical management port that provides out-of-band access to a Cisco device. Out-of-band access refers to access via a dedicated management channel that is used for device maintenance purposes only. The advantage of using a console port is that the device is accessible even if no networking services are configured e.g. performing the initial configuration. A computer running terminal emulation software and a special console cable to connect to the device are required for a console connection. |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SSH     | SSH is an in-band and recommended method for remotely establishing a secure CLI connection, through a virtual interface, over a network. Unlike a console connection, SSH connections require active networking services on the device, including an active interface configured with an address.                                                                                                                                                                                                                           |
| Telnet  | Telnet is an insecure, in-band method of remotely establishing a CLI session, through a virtual interface, over a network. Unlike SSH, Telnet does not provide a secure, encrypted connection and should only be used in a lab environment. User authentication, passwords, and commands are sent over the network in plaintext.                                                                                                                                                                                            |

**Note:** Some devices, such as routers, may also support a legacy auxiliary port that was used to establish a CLI session remotely over a telephone connection using a modem. Similar to a console connection, the AUX port is out-of-band and does not require networking services to be configured or available.

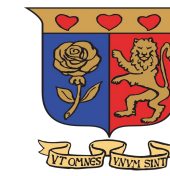
# Configure a Switch with Initial Settings: Switch Boot Sequence



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- After a Cisco switch is powered on, it goes through the following five-step boot sequence:
  - Step 1:** First, the switch loads a power-on self-test (POST) program stored in ROM. POST checks the CPU subsystem. It tests the CPU, DRAM, and the portion of the flash device that makes up the flash file system.
  - Step 2:** Next, the switch loads the boot loader software. The boot loader is a small program stored in ROM that is run immediately after POST successfully completes.
  - Step 3:** The boot loader performs low-level CPU initialization. It initializes the CPU registers, which control where physical memory is mapped, the quantity of memory, and its speed.
  - Step 4:** The boot loader initializes the flash file system on the system board.
  - Step 5:** Finally, the boot loader locates and loads a default IOS operating system software image into memory and gives control of the switch over to the IOS.

# Configure a Switch with Initial Settings: The boot system Command

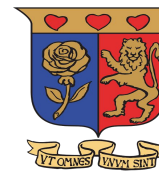


- The switch attempts to automatically boot by using information in the BOOT environment variable. If this variable is not set, the switch attempts to load and execute the first executable file it can find.
- The IOS operating system then initializes the interfaces using the Cisco IOS commands found in the startup-config file. The startup-config file is called **config.text** and is located in flash.
- In the example, the BOOT environment variable is set using the **boot system** global configuration mode command. Notice that the IOS is located in a distinct folder and the folder path is specified. Use the command **show boot** to see what the current IOS boot file is set to.

```
Sl(config)# boot system flash:/c2960-lanbasek9-mz.150-2.SE/c2960-lanbasek9-mz.150-2.SE.bin
```

| Command                                | Definition                  |
|----------------------------------------|-----------------------------|
| <b>boot system</b>                     | The main command            |
| <b>flash:</b>                          | The storage device          |
| <b>c2960-lanbasek9-mz.150-2.SE/</b>    | The path to the file system |
| <b>c2960-lanbasek9-mz.150-2.SE.bin</b> | The IOS file name           |

# Configure a Switch with Initial Settings: Switch LED Indicators



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**System LED (SYST):** Shows whether the system is receiving power and functioning properly.

**Redundant Power Supply LED (RPS):** Shows the RPS status.

**Port Status LED (STAT):** When green, indicates port status mode is selected, which is the default. Port status can then be understood by the light associated with each port.

**Port Duplex LED (DUPLX):** When green, indicates port duplex mode is selected. Port duplex can then be understood by the light associated with each port.

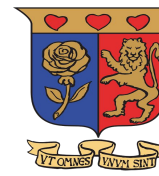
**Port Speed LED (SPEED):** When green, indicates port speed mode is selected. Port speed can then be understood by the light associated with each port.

**Power over Ethernet LED (PoE):** Present if the switch supports PoE. Indicates the PoE status of ports on the switch.

The Mode button is used to move between the different modes – STAT, DUPLX, SPEED, and PoE

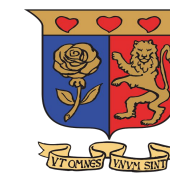


# Configure a Switch with Initial Settings: Switch LED Indicators (Cont.)



|                                                                                          | Off                     | Green       | Blinking Green           | Amber                        | Blinking Amber                          | Alternating Green/Amber  |
|------------------------------------------------------------------------------------------|-------------------------|-------------|--------------------------|------------------------------|-----------------------------------------|--------------------------|
| RPS                                                                                      | Off/No RPS              | RPS ready   | RPS up but not available | RPS standby or fault         | Internal PS failed, RPS providing power | N/A                      |
| PoE                                                                                      | Not selected, no issues | Selected    | N/A                      | N/A                          | Not selected, port issues present       | N/A                      |
| When the named mode is selected, the light associated with each physical port indicates: |                         |             |                          |                              |                                         |                          |
| STAT                                                                                     | No link or shutdown     | Link Up     | Activity                 | Port blocked preventing loop | Port blocked preventing loop            | Link fault               |
| DUPLEX                                                                                   | Half-duplex             | Full-duplex | N/A                      | N/A                          | N/A                                     | N/A                      |
| SPEED                                                                                    | 10Mbps                  | 100Mbps     | 1000Mbps                 | N/A                          | N/A                                     | N/A                      |
| PoE                                                                                      | PoE off                 | PoE on      | N/A                      | PoE disabled                 | PoE off due to fault                    | PoE denied (over budget) |

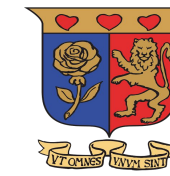
# Configure a Switch with Initial Settings: Recovering from a System Crash



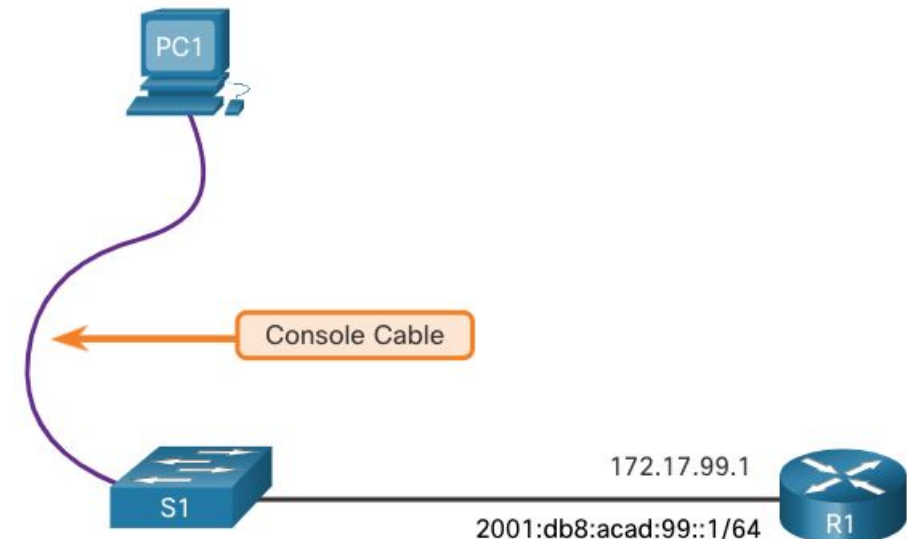
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- The boot loader provides access into the switch if the operating system cannot be used because of missing or damaged system files. The boot loader has a command line that provides access to the files stored in flash memory. The boot loader can be accessed through a console connection following these steps:
  - Step 1.** Connect a PC by console cable to the switch console port. Configure terminal emulation software to connect to the switch.
  - Step 2.** Unplug the switch power cord.
  - Step 3.** Reconnect the power cord to the switch and, within 15 seconds, press and hold down the **Mode** button while the System LED is still flashing green.
  - Step 4.** Continue pressing the **Mode** button until the System LED turns briefly amber and then solid green; then release the **Mode** button.
  - Step 5.** The boot loader **switch:** prompt appears in the terminal emulation software on the PC.
- The boot loader command line supports commands to format the flash file system, reinstall the operating system software, and recover a lost or forgotten password. For example, the **dir** command can be used to view a list of files within a specified directory.

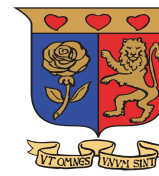
## Configure a Switch with Initial Settings: Switch Management Access



- To prepare a switch for remote management access, the switch must be configured with an IP address and a subnet mask.
  - To manage the switch from a remote network, the switch must be configured with a default gateway. This is very similar to configuring the IP address information on host devices.
  - In the figure, the switch virtual interface (SVI) on S1 should be assigned an IP address. The SVI is a virtual interface, not a physical port on the switch. A console cable is used to connect to a PC so that the switch can be initially configured.



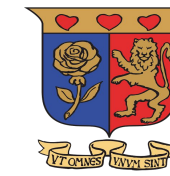
# Configure a Switch with Initial Settings: Switch SVI Configuration Example



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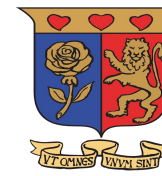
- By default, the switch is configured to have its management controlled through VLAN 1. All ports are assigned to VLAN 1 by default. For security purposes, it is considered a best practice to use a VLAN other than VLAN 1 for the management VLAN,  
**Step 1: Configure the Management Interface:** From VLAN interface configuration mode, an IPv4 address and subnet mask is applied to the management SVI of the switch.
- **Note:** The SVI for VLAN 99 will not appear as “up/up” until VLAN 99 is created and there is a device connected to a switch port associated with VLAN 99.
- **Note:** The switch may need to be configured for IPv6. For example, before you can configure IPv6 addressing on a Cisco Catalyst 2960 running IOS version 15.0, you will need to enter the global configuration command **sdm prefer dual-ipv4-and-ipv6 default** and then **reload** the switch.

# Configure a Switch with Initial Settings: Switch SVI Configuration Example (Cont.)



| Task                                             | IOS Commands                                                |
|--------------------------------------------------|-------------------------------------------------------------|
| Enter global configuration mode.                 | S1# <b>configure terminal</b>                               |
| Enter interface configuration mode for the SVI.  | S1(config)# <b>interface vlan 99</b>                        |
| Configure the management interface IPv4 address. | S1(config-if)# <b>ip address 172.17.99.11 255.255.255.0</b> |
| Configure the management interface IPv6 address  | S1(config-if)# <b>ipv6 address 2001:db8:acad:99::1/64</b>   |
| Enable the management interface.                 | S1(config-if)# <b>no shutdown</b>                           |
| Return to the privileged EXEC mode.              | S1(config-if)# <b>end</b>                                   |
| Save the running config to the startup config.   | S1# <b>copy running-config startup-config</b>               |

# Configure a Switch with Initial Settings: Switch SVI Configuration Example (Cont.)

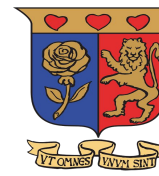


## •Step 2: Configure the Default Gateway

- The switch should be configured with a default gateway if it will be managed remotely from networks that are not directly connected.
- **Note:** Because, it will receive its default gateway information from a router advertisement (RA) message, the switch does not require an IPv6 default gateway.

| Task                                           | IOS Commands                                      |
|------------------------------------------------|---------------------------------------------------|
| Enter global configuration mode.               | S1# <b>configure terminal</b>                     |
| Configure the default gateway for the switch.  | S1(config)# <b>ip default-gateway 172.17.99.1</b> |
| Return to the privileged EXEC mode.            | S1(config-if)# <b>end</b>                         |
| Save the running config to the startup config. | S1# <b>copy running-config startup-config</b>     |

## Configure a Switch with Initial Settings: Switch SVI Configuration Example (Cont.)



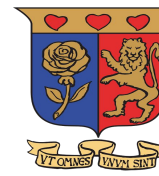
### •Step 3: Verify Configuration

- The **show ip interface brief** and **show ipv6 interface brief** commands are useful for determining the status of both physical and virtual interfaces. The output shown confirms that interface VLAN 99 has been configured with an IPv4 and IPv6 address.

**Note:** An IP address applied to the SVI is only for remote management access to the switch; this does not allow the switch to route Layer 3 packets.

```
S1# show ip interface brief
Interface      IP-Address      OK? Method      Status      Protocol
Vlan99         172.17.99.11    YES manual      down        down
(output omitted)
S1# show ipv6 interface brief
Vlan99          [down/down]
FE80::C27B:BCFF:FE4:A9C1
2001:DB8:ACAD:99::1
(output omitted)
```

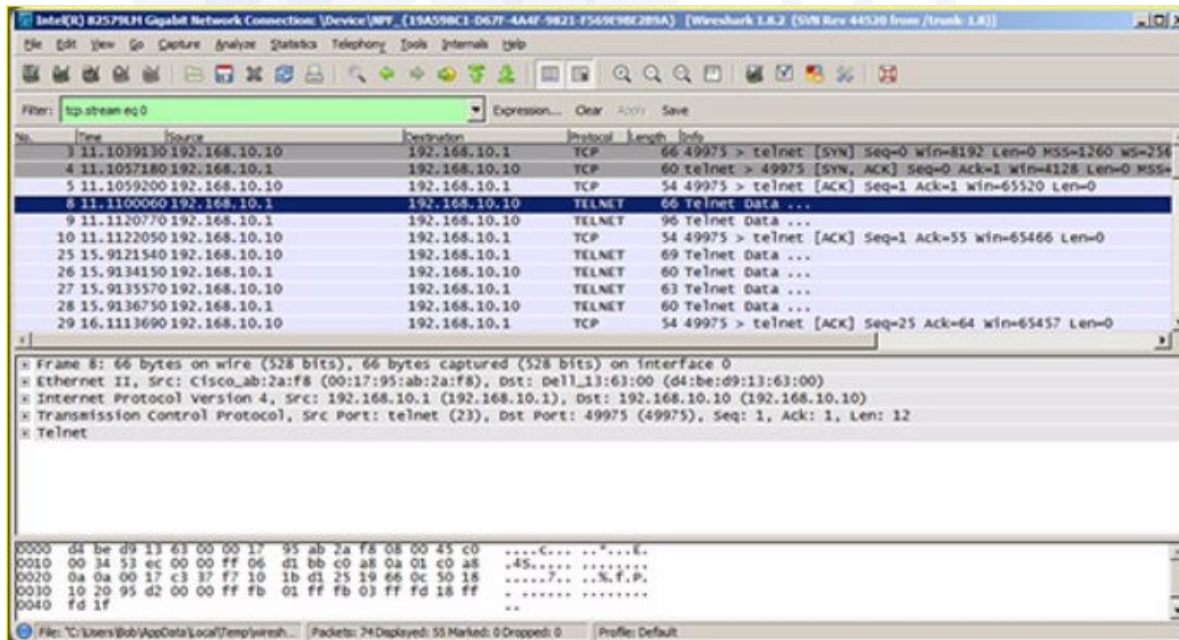
# Secure Remote Access: SSH Operation



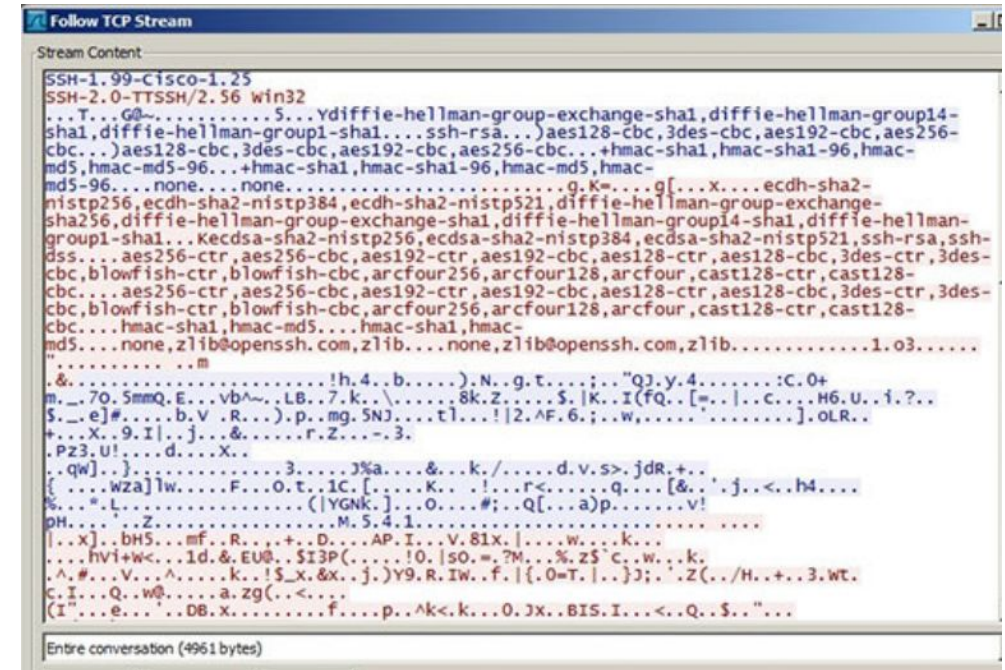
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- Secure Shell (SSH)
  - An alternative protocol to Telnet. Telnet uses unsecured plaintext of the username and password as well as the data transmitted.
  - SSH is more secure because it provides an encrypted management connection.

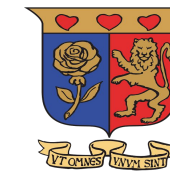
Wireshark Capture of Telnet



Wireshark Capture of SSH



## Secure Remote Access: SSH Operation (Cont.)



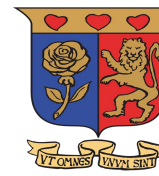
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- A switch must have an IOS version (k9 at the end of the IOS file name) that includes cryptographic capabilities in order to configure and use SSH.
  - Use the **show version** command to see the IOS version.



```
S1> show version
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M),
Version 15.0(2)SE, RELEASE SOFTWARE (fc1)
<output omitted>
```

# Secure Remote Access: Configuring SSH



1. Verify SSH support.
2. Configure the IP domain name.
3. Generate RSA key pairs.
4. Configure user authentication.
5. Configure the vty lines.
6. Enable SSH version 2.

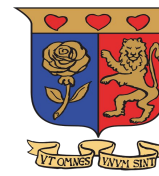
Commonly forgotten  
command that is used in key  
generation

```
S1# configure terminal
S1(config)# ip domain-name cisco.com
S1(config)# crypto key generate rsa
The name for the keys will be: S1.cisco.com
...
How many bits in the modulus [512]: 1024
...
S1(config)# username admin secret ccna
S1(config-line)# line vty 0 15
S1(config-line)# transport input ssh
S1(config-line)# login local
S1(config-line)# exit
S1(config)# ip ssh version 2
S1(config)# exit
S1#
```

The `login local` command  
forces the use of the local  
database for username/  
password.

Default is to accept both Telnet and  
SSH (transport input all)

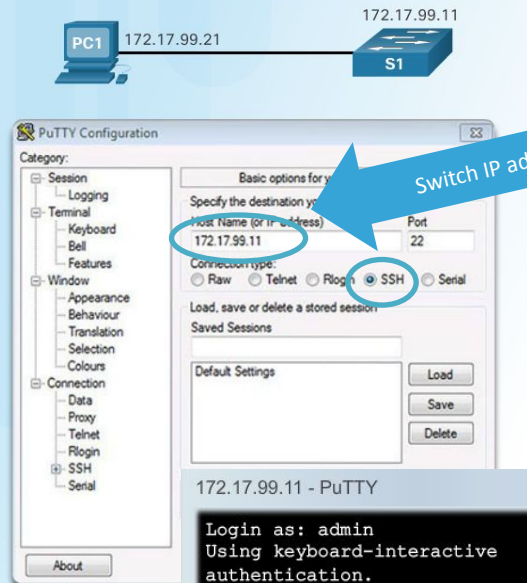
# Secure Remote Access: Verifying SSH



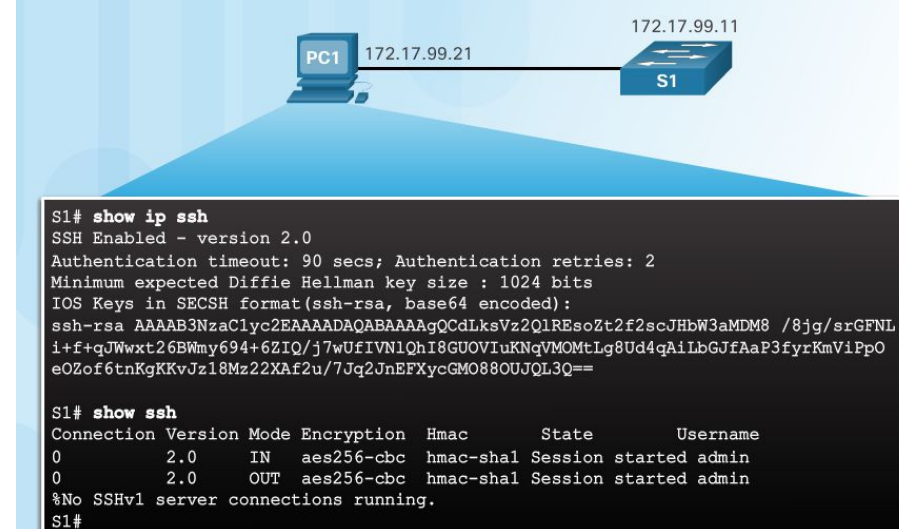
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- On the PC, connect to the switch using SSH.

## Configure PuTTY SSH Client Connection Parameters

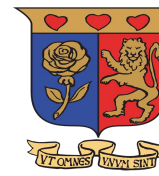


## Verify SSH Status and Settings



The PC is using SSH to communicate and issue commands on the switch.

# Switch Port Security: Secure Unused Ports



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The **interface range** command can be used to apply a configuration to several switch ports at one time.

## Disable Unused Ports



```
S1# show run
Building configuration...
```

```
...
version 15.0
hostname S1
```

```
...
interface FastEthernet0/4
 shutdown
```

```
!
interface FastEthernet0/5
 shutdown
```

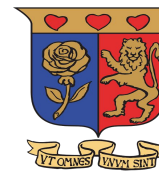
```
!
interface FastEthernet0/6
 description web server
```

```
!
interface FastEthernet0/7
 shutdown
```

```
!
```

```
...
```

Disable unused ports using the **shutdown** command.

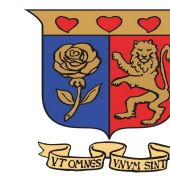


- Port security limits the number of valid MAC addresses allowed to transmit data through a switch port.
  - If a port has port security enabled and an unknown MAC address sends data, the switch presents a security violation.
  - Default number of secure MAC addresses allowed is 1.
- Methods use to configure MAC addresses within port security:
  - Static secure MAC addresses – manually configure  
**switchport port-security mac-address *mac-address***
  - Dynamic secure MAC addresses – dynamically learned and removed if the switch restarts
  - Sticky secure MAC addresses – dynamically learned and added to the running configuration (which can later be saved to the startup-config to permanently retain the MAC addresses)

**switchport port-security mac-address sticky *mac-address***

**Note:** Disabling sticky learning converts sticky MAC addresses to dynamic secure addresses and removes them from the running-config.

# Switch Port Security: Port Security: Violation Modes



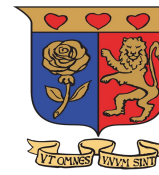
- Protect – data from unknown source MAC addresses are dropped; a security notification **IS NOT** presented by the switch
- Restrict - data from unknown source MAC addresses are dropped; a security notification **IS** presented by the switch and the violation counter increments.
- Shutdown – (default mode) interface becomes error-disabled and port LED turns off. The violation counter increments. Issues the shutdown and then the no shutdown command on the interface to bring it out of the error-disabled state.

| Violation Mode | Forwards Traffic | Sends Syslog Message | Displays Error Message | Increases Violation Counter | Shuts Down Port |
|----------------|------------------|----------------------|------------------------|-----------------------------|-----------------|
| Protect        | No               | No                   | No                     | No                          | No              |
| Restrict       | No               | Yes                  | No                     | Yes                         | No              |
| Shutdown       | No               | No                   | No                     | Yes                         | Yes             |

## Security Violations Occur In These Situations

- A station with MAC address that is not in the address table attempts to access the interface when the table is full.
- An address is being used on two secure interfaces in the same VLAN.

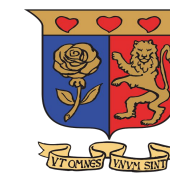
# Switch Port Security: Port Security: Configuring



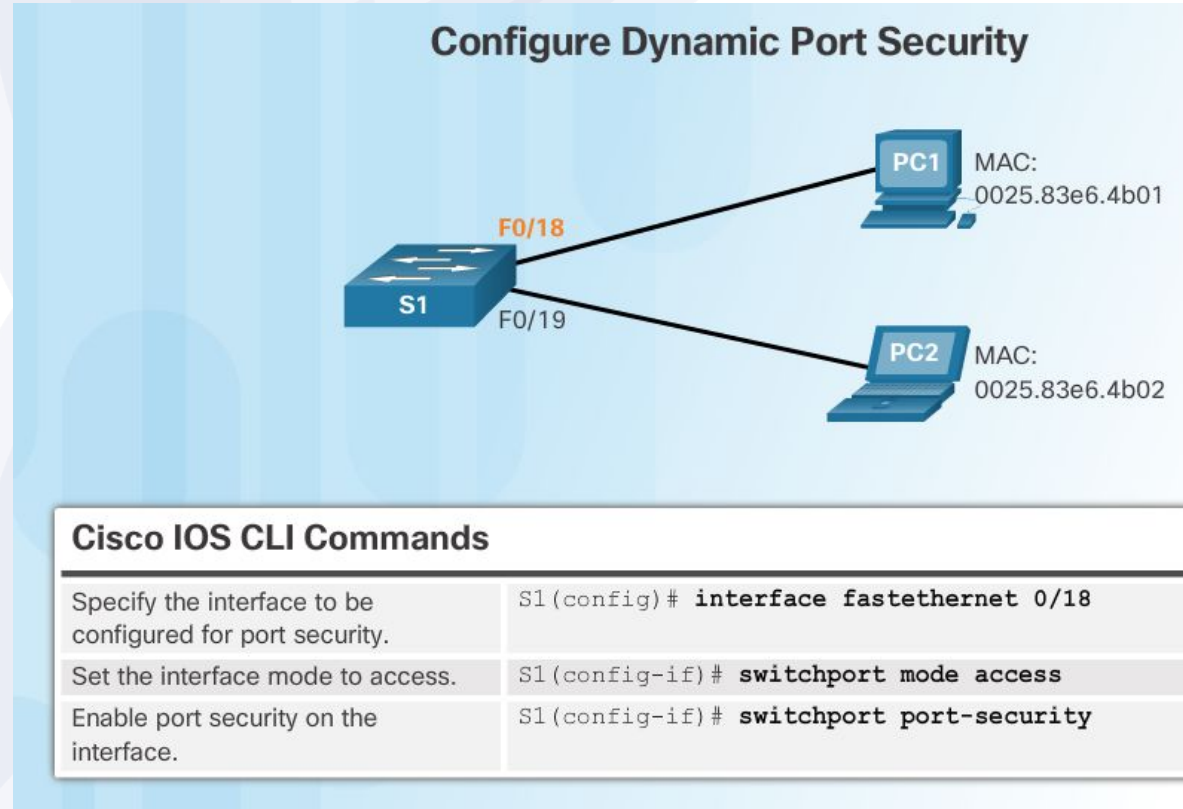
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| Feature                                | Default Setting                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------|
| Port security                          | Disabled on a port                                                                         |
| Maximum number of secure MAC addresses | 1                                                                                          |
| Violation mode                         | Shutdown. The port shuts down when the maximum number of secure MAC addresses is exceeded. |
| Sticky address learning                | Disabled                                                                                   |

# Switch Port Security: Port Security: Configuring (Cont.)

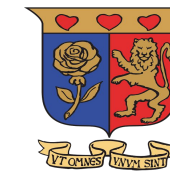


- Before configuring port-security features, place the port in access mode and use the **switchport port-security** interface configuration command to enable port security on an interface.



Most common configuration error is to forget this command!

# Switch Port Security: Port Security: Configuring (Cont.)



## Configure Sticky Port Security

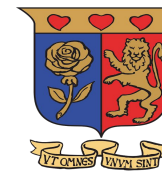


### Cisco IOS CLI Commands

|                                                                 |                                                                         |
|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| Specify the interface to be configured for port security.       | <code>S1(config)# interface fastethernet 0/19</code>                    |
| Set the interface mode to access.                               | <code>S1(config-if)# switchport mode access</code>                      |
| Enable port security on the interface.                          | <code>S1(config-if)# switchport port-security</code>                    |
| Set the maximum number of secure addresses allowed on the port. | <code>S1(config-if)# switchport port-security maximum 10</code>         |
| Enable sticky learning.                                         | <code>S1(config-if)# switchport port-security mac-address sticky</code> |

Most common configuration error is to forget this command!

# Switch Port Security: Port Security: Verifying



- Use the **show port-security interface** command to verify the maximum number of MAC addresses allowed on a particular port and how many of those addresses were learned dynamically using sticky.

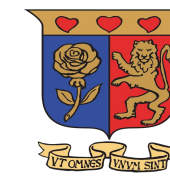
## Dynamic

```
S1# show port-security interface fastethernet 0/18
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 0025.83e6.4b01:1
Security Violation Count : 0
```

## Sticky

```
S1# show port-security interface fastethernet 0/19
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 10
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 0025.83e6.4b02:1
Security Violation Count : 0
```

# Switch Port Security: Port Security: Verifying (Cont.)



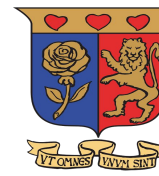
- Use the **show running-config** command to see learned MAC addresses added to the configuration.

```
S1# show run | begin FastEthernet 0/19
interface FastEthernet0/19
  switchport mode access
  switchport port-security maximum 10
  switchport port-security
  switchport port-security mac-address sticky
  switchport port-security mac-address sticky 0025.83e6.4b02
```

- The **show port-security address** command shows how MAC addresses were learned on a particular port.

```
S1# show port-security address
Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
(mins)
----
1       0025.83e6.4b01   SecureDynamic       Fa0/18   -
1       0025.83e6.4b02   SecureSticky        Fa0/19   -
-----
```

# Switch Port Security: Ports in Error Disabled State



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- Switch console messages display when a port security violation occurs. Notice the port link status changes to down.

```
Sep 20 06:44:54.966: %PM-4-ERR_DISABLE: psecure-violation error detected on Fa0/18,  
putting Fa0/18 in err-disable state  
Sep 20 06:44:54.966: %PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred,  
caused by MAC address 000c.292b.4c75 on port FastEthernet0/18.  
Sep 20 06:44:55.973: %LINEPROTO-5-PPDOWN: Line protocol on Interface  
FastEthernet0/18, changed state to down  
Sep 20 06:44:56.971: %LINK-3-UPDOWN: Interface FastEthernet0/18, changed state to down
```

# Switch Port Security: Ports in Error Disabled State (Cont.)



- Check the port status and the port security settings.

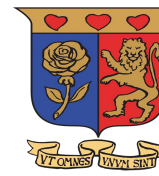
```
S1# show interface fa0/18 status
Port Name  Status      Vlan  Duplex  Speed  Type
Fa0/18     err-disabled 1     auto   auto   10/100BaseTX

S1# show port-security interface fastethernet 0/18
Port Security           : Enabled
Port Status             : Secure-shutdown
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type               : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses    : 1
Total MAC Addresses      : 0
Configured MAC Addresses : 0
Sticky MAC Addresses     : 0
Last Source Address:Vlan : 000c.292b.4c75:1
Security Violation Count : 1
```

- Do not re-enable a port until the security threat is investigated and eliminated.
- Notice that you must first shut the port down and then issue the **no shutdown** command in order to use the particular port again after a security violation has occurred

```
S1(config)# interface FastEthernet 0/18
S1(config-if)# shutdown
Sep 20 06:57:28.532: %LINK-5-CHANGED: Interface FastEthernet0/18, changed state to
administratively down
S1(config-if)# no shutdown
Sep 20 06:57:48.186: %LINK-3-UPDOWN: Interface FastEthernet0/18, changed state to up
Sep 20 06:57:49.193: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/18, changed state to up
```

# Configure Switch Ports: Duplex Communication



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- Full-duplex communication increases bandwidth efficiency by allowing both ends of a connection to transmit and receive data simultaneously. This is also known as bidirectional communication, and it requires microsegmentation.
- A microsegmented LAN is created when a switch port has only one device connected and is operating in full-duplex mode. There is no collision domain associated with a switch port operating in full-duplex mode.
- Unlike full-duplex communication, half-duplex communication is unidirectional. Half-duplex communication creates performance issues because data can flow in only one direction at a time, often resulting in collisions.
- Gigabit Ethernet and 10 Gb NICs require full-duplex connections to operate. In full-duplex mode, the collision detection circuit on the NIC is disabled. Full-duplex offers 100 percent efficiency in both directions (transmitting and receiving). This results in a doubling of the potential use of the stated bandwidth.

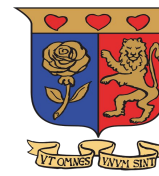
# Configure Switch Ports: Configure Switch Ports at the Physical Layer



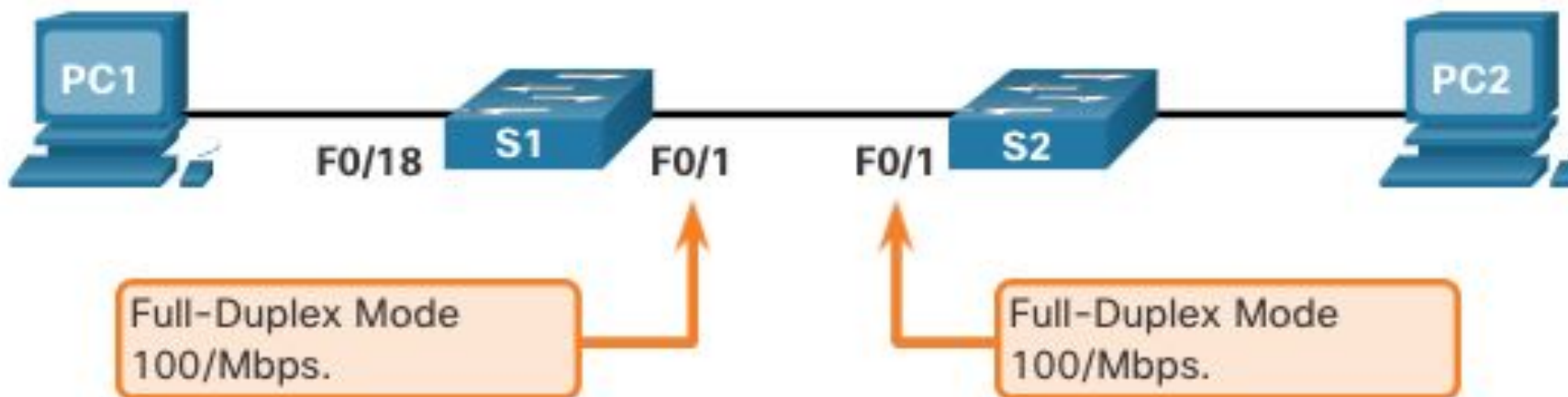
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- Switch ports can be manually configured with specific duplex and speed settings. The respective interface configuration commands are **duplex** and **speed**.
- The default setting for both duplex and speed for switch ports on Cisco Catalyst 2960 and 3560 switches is auto. The 10/100/1000 ports operate in either half- or full-duplex mode when they are set to 10 or 100 Mbps and operate only in full-duplex mode when it is set to 1000 Mbps (1 Gbps).
- Autonegotiation is useful when the speed and duplex settings of the device connecting to the port are unknown or may change. When connecting to known devices such as servers, dedicated workstations, or network devices, a best practice is to manually set the speed and duplex settings.
- When troubleshooting switch port issues, it is important that the duplex and speed settings are checked.
- **Note:** Mismatched settings for the duplex mode and speed of switch ports can cause connectivity issues. Autonegotiation failure creates mismatched settings.
- All fiber-optic ports, such as 1000BASE-SX ports, operate only at one preset speed and are **always full-duplex**.

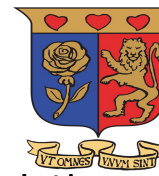
# Configure Switch Ports: Configure Switch Ports at the Physical Layer (Cont.)



| Task                                           | IOS Commands                                  |
|------------------------------------------------|-----------------------------------------------|
| Enter global configuration mode.               | S1# <b>configure terminal</b>                 |
| Enter interface configuration mode.            | S1(config)# <b>interface FastEthernet 0/1</b> |
| Configure the interface duplex.                | S1(config-if)# <b>duplex full</b>             |
| Configure the interface speed.                 | S1(config-if)# <b>speed 100</b>               |
| Return to the privileged EXEC mode.            | S1(config-if)# <b>end</b>                     |
| Save the running config to the startup config. | S1# <b>copy running-config startup-config</b> |

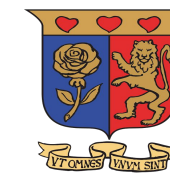


# Configure Switch Ports: Auto-MDIX



- When automatic medium-dependent interface crossover (auto-MDIX) is enabled, the switch interface automatically detects the required cable connection type (straight-through or crossover) and configures the connection appropriately.
- When connecting to switches without the auto-MDIX feature, straight-through cables must be used to connect to devices such as servers, workstations, or routers. Crossover cables must be used to connect to other switches or repeaters.
- With auto-MDIX enabled, either type of cable can be used to connect to other devices, and the interface automatically adjusts to communicate successfully.
- On newer Cisco switches, the **mdix auto** interface configuration mode command enables the feature. When using auto-MDIX on an interface, the interface speed and duplex must be set to **auto** so that the feature operates correctly.
- **Note:** The auto-MDIX feature is enabled by default on Catalyst 2960 and Catalyst 3560 switches but is not available on the older Catalyst 2950 and Catalyst 3550 switches.
- To examine the auto-MDIX setting for a specific interface, use the **show controllers ethernet-controller** command with the **phy** keyword. To limit the output to lines referencing auto-MDIX, use the **include Auto-MDIX** filter.

# Configure Switch Ports: Switch Verification Commands



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| Task                                         | IOS Commands                                                                                                         |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Display interface status and configuration.  | S1# <b>show interfaces</b> [ <i>interface-id</i> ]                                                                   |
| Display current startup configuration.       | S1# <b>show startup-config</b>                                                                                       |
| Display current running configuration.       | S1# <b>show running-config</b>                                                                                       |
| Display information about flash file system. | S1# <b>show flash</b>                                                                                                |
| Display system hardware and software status. | S1# <b>show version</b>                                                                                              |
| Display history of command entered.          | S1# <b>show history</b>                                                                                              |
| Display IP information about an interface.   | S1# <b>show ip interface</b> [ <i>interface-id</i> ]<br>OR<br>S1# <b>show ipv6 interface</b> [ <i>interface-id</i> ] |
| Display the MAC address table.               | S1# <b>show mac-address-table</b><br>OR<br>S1# <b>show mac address-table</b>                                         |

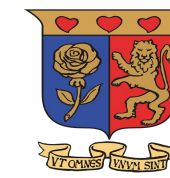
# Configure Switch Ports: Verify Switch Port Configuration



- The **show running-config** command can be used to verify that the switch has been correctly configured. From the sample abbreviated output on S1, some important information is shown in the figure:
  - Fast Ethernet 0/18 interface configured with the management VLAN 99
  - VLAN 99 configured with an IPv4 address of 172.17.99.11 255.255.255.0
  - Default gateway set to 172.17.99.1

```
S1# show running-config
Building configuration...
Current configuration : 1466 bytes
!
(output omitted)
interface Vlan99
  ip address 172.17.99.11 255.255.255.0
  ipv6 address 2001:DB8:ACAD:99::1/64
!
ip default-gateway 172.17.99.1
```

# Configure Switch Ports: Verify Switch Port Configuration (Cont.)

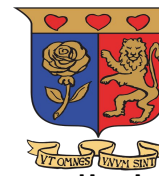


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- The **show interfaces** command is another commonly used command, which displays status and statistics information on the network interfaces of the switch. The **show interfaces** command is frequently used when configuring and monitoring network devices.
- The first line of the output for the **show interfaces fastEthernet 0/18** command indicates that the FastEthernet 0/18 interface is up/up, meaning that it is operational. Further down, the output shows that the duplex is full and the speed is 100 Mbps.

```
S1# show interfaces fastEthernet 0/18
FastEthernet0/18 is up, line protocol is up (connected)
  Hardware is Fast Ethernet, address is 0025.83e6.9092 (bia 0025.83e6.9092)
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is 10/100BaseTX
```

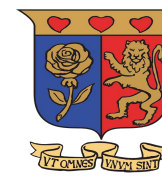
# Configure Switch Ports: Network Access Layer Issues



- The output from the **show interfaces** command is useful for detecting common media issues. One of the most important parts of this output is the display of the line and data link protocol status, as shown in the example.
- The first parameter (FastEthernet0/18 is up) refers to the hardware layer and indicates whether the interface is receiving a carrier detect signal. The second parameter (line protocol is up) refers to the data link layer and indicates whether the data link layer protocol keepalives are being received. Based on the output of the **show interfaces** command, possible problems can be fixed as follows:
  - If the interface is up and the line protocol is down, a problem exists. There could be an encapsulation type mismatch, the interface on the other end could be error-disabled, or there could be a hardware problem.
  - If the line protocol and the interface are both down, a cable is not attached, or some other interface problem exists. For example, in a back-to-back connection, the other end of the connection may be administratively down.
  - If the interface is administratively down, it has been manually disabled (the **shutdown** command has been issued) in the active configuration.

```
SI# show interfaces fastEthernet 0/18
FastEthernet0/18 is up, line protocol is up (connected)
Hardware is Fast Ethernet, address is 0025.83e6.9092 (bia 0025.83e6.9092) MTU 1500 bytes, BW
100000 Kbit/sec, DLY 100 usec,
```

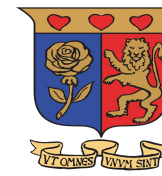
# Configure Switch Ports: Network Access Layer Issues (Cont.)



- The **show interfaces** command output displays counters and statistics for the FastEthernet0/18 interface, as shown here:

```
S1# show interfaces fastEthernet 0/18
FastEthernet0/18 is up, line protocol is up (connected)
  Hardware is Fast Ethernet, address is 0025.83e6.9092 (bia 0025.83e6.9092)
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is 10/100BaseTX
  input flow-control is off, output flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input never, output 00:00:01, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    2295197 packets input, 305539992 bytes, 0 no buffer
    Received 1925500 broadcasts (74 multicasts)
      0 runs, 0 giants, 0 throttles
      3 input errors, 3 CRC, 0 frame, 0 overrun, 0 ignored
      0 watchdog, 74 multicast, 0 pause input
      0 input packets with dribble condition detected
    3594664 packets output, 436549843 bytes, 0 underruns
      8 output errors, 1790 collisions, 10 interface resets
      0 unknown protocol drops
      0 babbles, 235 late collision, 0 deferred
```

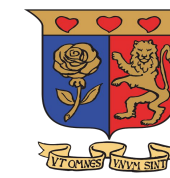
# Configure Switch Ports: Network Access Layer Issues (Cont.)



| Error Type             | Description                                                                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Input Errors</b>    | Total number of errors. It includes runts, giants, no buffer, CRC, frame, overrun, and ignored counts.                                                                              |
| <b>Runts</b>           | Packets that are discarded because they are smaller than the minimum packet size for the medium. For instance, any Ethernet packet that is less than 64 bytes is considered a runt. |
| <b>Giants</b>          | Packets that are discarded because they exceed the maximum packet size for the medium. For example, any Ethernet packet that is greater than 1,518 bytes is considered a giant.     |
| <b>CRC</b>             | CRC errors are generated when the calculated checksum is not the same as the checksum received.                                                                                     |
| <b>Output Errors</b>   | Sum of all errors that prevented the final transmission of datagrams out of the interface that is being examined.                                                                   |
| <b>Collisions</b>      | Number of messages retransmitted because of an Ethernet collision.                                                                                                                  |
| <b>Late Collisions</b> | A collision that occurs after 512 bits of the frame have been transmitted                                                                                                           |

Some media errors are not severe enough to cause the circuit to fail but do cause network performance issues. The table explains some of these common errors which can be detected using the **show interfaces** command.

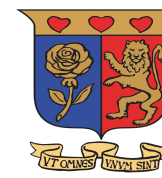
# Configure Switch Ports: Interface Input and Output Errors



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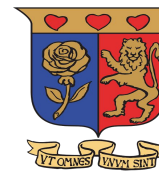
- “Input errors” is the sum of all errors in datagrams that were received on the interface being examined. This includes runts, giants, CRC, no buffer, frame, overrun, and ignored counts. The reported input errors from the **show interfaces** command include the following:
- **Runt Frames** - Ethernet frames that are shorter than the 64-byte minimum allowed length are called runts. Malfunctioning NICs are the usual cause of excessive runt frames, but they can also be caused by collisions.
- **Giants** - Ethernet frames that are larger than the maximum allowed size are called giants.
- **CRC errors** - On Ethernet and serial interfaces, CRC errors usually indicate a media or cable error. Common causes include electrical interference, loose or damaged connections, or incorrect cabling. If you see many CRC errors, there is too much noise on the link and you should inspect the cable. You should also search for and eliminate noise

# Configure Switch Ports: Interface Input and Output Errors (Cont.)

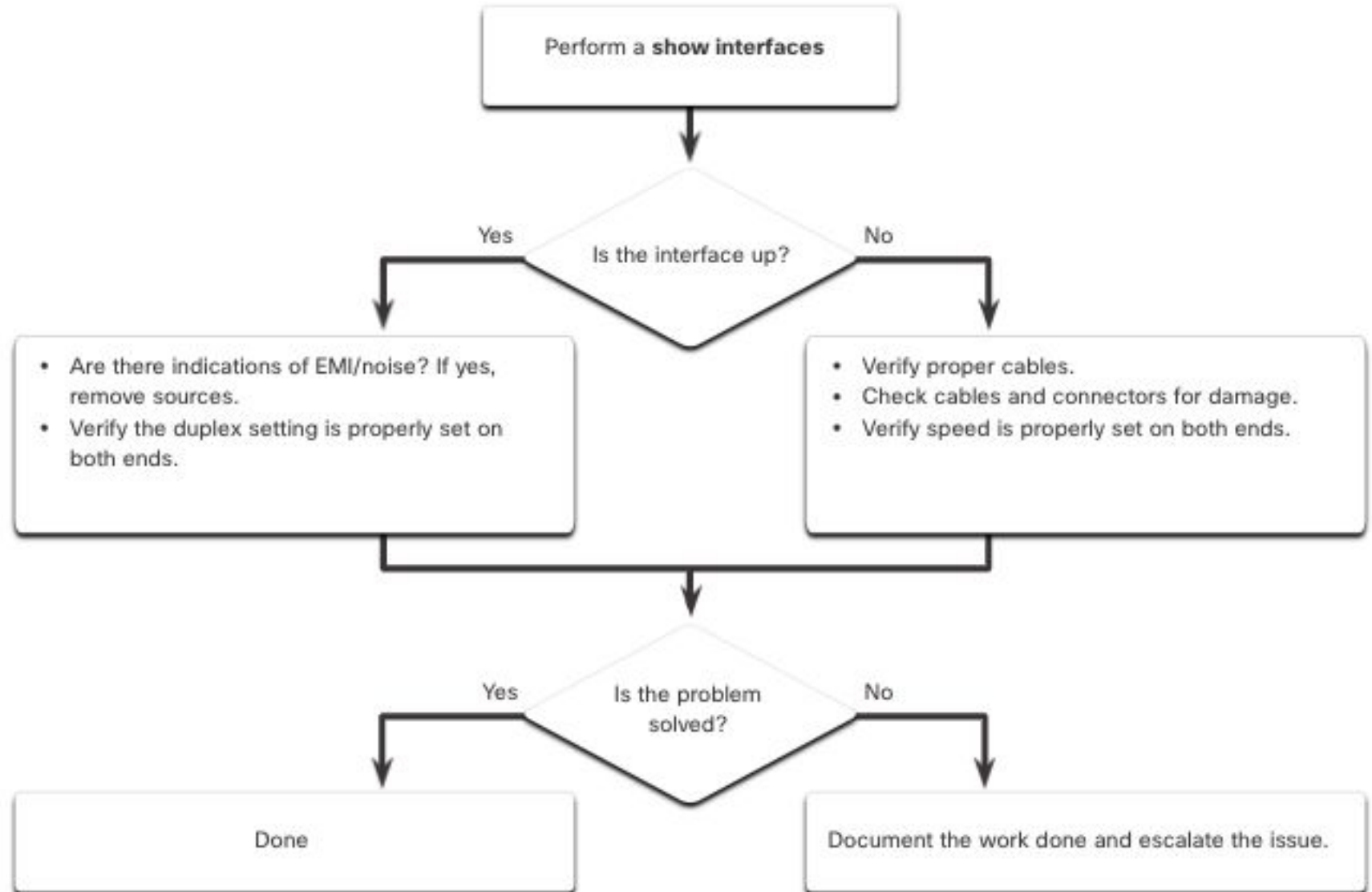


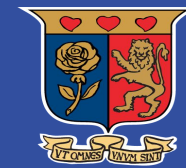
- “Output errors” is the sum of all errors that prevented the final transmission of datagrams out the interface that is being examined. The reported output errors from the **show interfaces** command include the following:
  - **Collisions** - Collisions in half-duplex operations are normal. However, you should never see collisions on an interface configured for full-duplex communication.
  - **Late collisions** - A late collision refers to a collision that occurs after 512 bits of the frame have been transmitted. Excessive cable lengths are the most common cause of late collisions. Another common cause is duplex misconfiguration.

# Configure Switch Ports: Troubleshooting Network Access Layer Issues



To troubleshoot scenarios involving no connection, or a bad connection, between a switch and another device, follow the general process shown in the figure.





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END